

Vera Zhou (Yuxin Zhou)

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EDUCATION

Carnegie Mellon University — Dietrich College of Humanities & Social Sciences

Pittsburgh, PA

B.S. in Statistics and Machine Learning

Aug 2023 – May 2027

Relevant Coursework: Introduction to Machine Learning, Natural Language Processing, Modern Regression, Statistical Computing, Principles of Imperative Computation, Probability & Statistical Inference.

RESEARCH EXPERIENCE

Cross-Expertise Communication Clarity

May 2026 – Aug 2026

Advisor: Dr. Pingbo Tang, Carnegie Mellon University

- Designed a 24-item binary clarity rubric across four sub-dimensions (Structuredness, Term Grounding, Focus, Resolution) to measure how effectively technical explanations transfer usable information across complementary-expertise boundaries, adapting items from validated instruments (PEMAT, CDC Clear Communication Index).
- Built a ~40-passage annotated calibration dataset and ran a dual-track validation of human raters against three LLM judges (Claude, DeepSeek, Gemini), quantifying reliability with Krippendorff's α and analyzing disagreement by sub-dimension, passage type, and linguistic feature.
- Outcome:** Delivered a reusable measurement pipeline and technical report feeding a downstream information-retrieval system, validated on a real vendor–factory diagnostic transcript (Roll-Kraft × DMI).

Cross-Cultural Semantics via an LLM Agent Framework

Jan 2026 – Jul 2026

Advisor: Dr. Wei Li, Chinese Academy of Sciences

- Conducted a faithful, cost-bounded reproduction of Agonistic Corpus Synthesis (ACS), a multi-agent debate framework for classical Chinese poetry translation, on the PoetMT benchmark, running intrinsic-quality and downstream fine-tuning experiments (gpt-4o-mini as agent/judge; Qwen2.5-0.5B + LoRA as learner).
- Diagnosed an asymmetric reproducibility profile (intrinsic claims partially replicate; downstream claims invert) as structural rather than incidental, and identified same-family single-judge self-preference bias as the core methodological weakness.
- Designed Multi-Judge Council ACS (MJC-ACS), a PoLL-inspired extension with a heterogeneous judge panel (GPT-4o / Claude 3.5 Sonnet / Qwen2.5-72B) and calibrated-weighted aggregation.
- Outcome:** Delivered a pre-registered evaluation protocol with literature-grounded projected results and authored a full research paper independently, accepted for oral presentation at the 2nd International Conference on AI-enabled Education (AIEE 2026).

PROFESSIONAL EXPERIENCE

Whale Cloud Technology Co., Ltd.

Nanjing, Jiangsu

Robotic Arm Control & Embodied AI Intern

May 2026 – Aug 2026

- Real-Robot Control & Motion Planning:** Built a ROS 2 + MoveIt 2 control environment from scratch for the 5-DoF SO-101 arm, establishing a full pipeline from coordinate input to inverse kinematics, joint-space planning, and real-robot execution; resolved the over-constrained IK problem, controller crashes, and ROS network interference.
- Teleoperation Development:** Developed and integrated multiple teleoperation systems (VR headset, leader-follower arm mapping, Joy-Con control), implementing coordinate-frame alignment, IK joint-jump suppression, and gripper calibration for stable, safe operation.
- VLA Model Training:** Ran the full Vision-Language-Action training workflow on SmolVLA and Lingbot-VLA, collecting and validating datasets and diagnosing inference-stage camera frame-drop issues (15–20 vs. 30 fps target).

Machine Learning Department, Carnegie Mellon University

Pittsburgh, PA

Peer Tutor, Introduction to Machine Learning (10-301)

Oct 2025 – Dec 2025

- Prepared weekly review materials and led study sessions on supervised learning, neural networks, RNNs, Transformers, and introductory RL, and guided students in debugging experiment code — earning 95%+ satisfaction in anonymous feedback.

Chengdu Heneng Chuangyue Software Co., Ltd.

Chengdu, Sichuan

Data Annotation Specialist

Apr 2023 – May 2023

- Independently annotated 5,000+ surveillance images with 98% accuracy for a smart-city detection system, raising downstream object-detection accuracy by ~6%; optimized the workflow via batching, templated labels, and file-naming automation to improve efficiency by 25%.

PROJECTS

SlotSync — Campus Study-Partner & Group Scheduling Platform

Fall 2025

Backend / Algorithm Lead, HackCMU (24h Hackathon, 3-person team)

- Designed and implemented friend-matching and multi-user common-availability algorithms (weighted by course overlap, major, and activity) achieving near-instant matching, and delivered API-driven, explainable When2Meet-style recommendations with real-time updates.
- Outcome:** Completed a full-feature prototype (status board, smart matching, group scheduling) with a stable live demo praised by the mentor panel.

Model Implementation & Analysis — Introduction to Machine Learning (10-301)

Fall 2025

- Implemented core ML models (k-NN, Logistic Regression, Neural Networks, CNN) and an RNN language model, then ran Policy-Gradient RL on Atari Pong, training an agent from random actions to stable positive returns.

TECHNICAL SKILLS

Programming & Tools: Python, R, SQL; Pandas, NumPy, PyTorch, Matplotlib; Excel, Word, PowerPoint.

Languages: Chinese (Native) · English (Fluent, TOEFL 117/120).